Foundations of Machine Learning: Cardiac MRI Denoising with Restormer

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***Abstract*—Cardiac Magnetic Resonance Imaging (MRI) is a vital tool for assessing cardiac anatomy; however, acquiring high-quality imaging relies on expensive high-field scanners. This cost creates an accessibility gap for smaller clinics that depend on lower-field systems, where images often have lower signal-to-noise ratio (SNR). This project investigates a deep learning-based denoising approach to improve cardiac MRI image quality while preserving anatomical structure.**

To address the lack of large paired noisy/clean cardiac MRI datasets [1], [2], we train a restoration-oriented Transformer (Restormer) using a synthetic data pipeline that injects randomized Gaussian and Rician noise into grayscale images. On the synthetic-noise validation set, the trained model achieves an average PSNR of 33.66 dB and SSIM of 0.93 at epoch 100, indicating stable convergence under constrained training resources.

In addition, benchmark using a Sunnybrook cardiac MRI images (reference image + synthetic noise) shows that the proposed model outperforms BM3D on that example, achieving 36.98 dB/0.89 versus BM3D's 32.38 dB/0.69. Qualitatively, the model suppresses noise while preserving fine anatomical textures that are often over-smoothed by classical filtering.

Key lessons learned highlight that mixed Gaussian/Rician synthetic noise modeling is a practical substitute when paired clinical data are scarce. Future work will extend the method to 3D volumetric processing and evaluate generalization using raw k-space data and multi-center clinical datasets.

***Index Terms*—Cardiac MRI, Denoising, Restormer, Deep Learning, Rician Noise.**

I. Introduction

A. Context and Background Knowledge

Cardiac magnetic resonance imaging (MRI) is a critical tool for assessing cardiac anatomy and function. However, the quality of these images is often compromised by acquisition noise and limited scan times. Noise weakens structural boundaries and obscures small details, which reduces clinical interpretability and adversely affects downstream analysis. MRI denoising aims to remove this noise while rigorously preserving the underlying anatomical structure.

B. Problem Statement and Research Gap

This work is motivated by a distinct research gap formed by two practical constraints in medical imaging.

First, strictly limiting the budget for hardware creates an "imaging gap." Modern radiology departments rely on high-field scanners (1.5T, 3T, or 7T) to combat noise and aliasing caused by accelerating scans for high temporal resolution [3]. However, these machines represent a massive financial investment (often >3 million USD). Smaller clinics rely on lower-field scanners (0.05T to 1.0T) which are accessible but produce images with significantly lower Signal-to-Noise Ratios (SNR) [3].

Second, existing denoising standards are insufficient for these lower-quality inputs. BM3D has served as the industry standard for non-learning denoising [4]. However, BM3D assumes Gaussian noise distributions and often performs poorly on Rician noise common in MRI. Furthermore, it tends to over-smooth images when noise levels are high, removing fine structures critical for diagnosis [5]. Access to large-scale biobank datasets to train better models is also restricted by high subscription costs [6], [7].

C. Summary of Results and Novelty

Results: The proposed Restormer model reaches an average validation performance of **33.66 dB** PSNR and **0.93** SSIM at epoch 100 under the synthetic-noise evaluation protocol. Separately, in a paired Sunnybrook case study (reference image with synthetically added noise), the proposed model achieves **36.98 dB/0.89**, exceeding BM3D's **32.38 dB/0.69** on the same example.

Novelty: The novelty of this work lies in adapting Restormer—originally designed for general image restoration—to cardiac MRI-like noise characteristics by using a synthetic noise generation pipeline that combines Gaussian and Rician corruption. This design targets MRI magnitude-image noise behavior without requiring large paired clinical datasets, and it provides a practical path to deployable denoising models under limited data and hardware constraints.

D. Impact on Theory and Practice

Impact on Theory: This study validates that Transformer-based architectures with linear attention complexity can model cardiac structures without the quadratic computational cost of standard ViTs.

Impact on Practice: This solution offers a method to improve image quality algorithmically rather than physically. It has the potential to enable small clinics using low-field scanners to produce improved images without requiring expensive hardware investments.

E. Report Structure

The remainder of this report is structured as follows:

- Section II reviews related work in medical image denoising and transformer architectures.
- Section III details the methodology, including the Restormer architecture, data simulation pipeline, and training protocols.
- Section IV presents the experimental setup, quantitative results, and qualitative visual comparisons.
- Section V concludes with findings, lessons learnt, and future directions.

II. Related Work

Cardiac MRI denoising aims to reduce noise while keeping tissue boundaries and fine anatomical texture. A core difficulty is that MRI noise characteristics depend on the acquisition and reconstruction pipeline. In many settings, noise can be approximated as Gaussian in complex-valued measurements, while magnitude images can exhibit Rician-like behavior, especially in low-signal regions [8], [9]. This section reviews classical denoising, CNN-based approaches, and Transformer-based restoration, then summarizes the gap addressed in this report.

A. Classical Approaches

Patch-based collaborative filtering methods such as BM3D group similar 2D patches into 3D stacks and apply transform-domain shrinkage [4], [10], [11]. BM3D is effective under additive white Gaussian noise (AWGN), but magnitude MRI can deviate from AWGN due to Rician-like distributions [5]. When the assumed noise model differs from the true noise, classical methods may produce biased estimates in low-signal regions and may attenuate subtle textures. These methods are not data-adaptive: the algorithmic rules remain fixed across imaging protocols and patient-specific anatomy.

B. Convolutional Neural Networks

CNN denoisers learn a mapping from noisy to clean images from paired data [12]. In medical imaging, U-Net-style encoder–decoder models are widely used because skip connections transfer high-resolution spatial information to the decoder, reducing loss of fine details during downsampling [13]. CNNs can learn anatomical priors from data, but may be sensitive to distribution shift when noise type, noise level, or imaging protocol differs from training.

C. Transformers and Attention for Image Restoration

Transformers provide global context modeling through self-attention [14]. A direct Vision Transformer formulation tokenizes images into patches [15], but full self-attention scales quadratically with the number of tokens, which is expensive for high-resolution restoration [16].

D. The Restormer Architecture

Restormer is a Transformer designed for image restoration with improved efficiency [16]. It replaces spatial attention with channel-wise attention through Multi-Dconv Head Transposed Attention (MDTA). It also uses a Gated-Dconv Feed-Forward Network (GDFN) to modulate features and suppress less useful responses during restoration.

E. Research Gap and Reason for Model Choosing

For cardiac MRI magnitude images, noise can be closer to Gaussian in some settings and Rician-like in others [8], so a single AWGN-only assumption can be insufficient. This report addresses the gap by training Restormer using mixed Gaussian and Rician noise injection to better match MRI noise behavior in practice, while keeping training feasible on limited hardware.

III. Methods

This section details the research objectives, the theoretical formulation of the denoising problem, the Restormer implementation, the data simulation strategy, and the training protocols.

A. Research Objectives

To address the research gap identified in Section I, this study pursues the following objectives:

RO1: Architecture Development. Implement a Restormer-based U-Net capable of processing cardiac MRI inputs.
Significance: Validates linear-complexity attention ($O(N)$) for medical imaging compared with quadratic attention ($O(N^2)$).

RO2: Synthetic Data Training. Simulate realistic MRI noise profiles with Gaussian and Rician noise on open-source datasets (13.5k images).
Significance: Establishes a cost-effective strategy when paired clinical data are scarce or expensive.

RO3: Performance Optimization. Tune hyperparameters (learning rate, batch size, network depth) to maximize PSNR/SSIM.
Significance: Ensures convergence within hardware constraints while improving restoration quality.

RO4: Benchmarking. Compare against a classical baseline (BM3D).
Significance: Provides empirical evidence for practical adoption relative to established filtering methods.

B. Research Questions and Hypotheses

RQ1
Does a Restormer denoiser trained with mixed Gaussian and Rician noise improve PSNR/SSIM compared with BM3D under the same evaluation protocol?

H1
Restormer achieves higher PSNR and SSIM than BM3D under the same evaluation.

RQ2
Does mixed-noise training improve robustness to real cardiac MRI appearance compared with a single-noise assumption?
H2
Mixed-noise training improves qualitative structure preservation (edges/textures), especially in low-signal regions.

C. Problem Formulation

Let $y \in \mathbb{R}^{H\times W\times C}$ denote the observed noisy image and $x \in \mathbb{R}^{H\times W\times C}$ the latent clean image:

$$y = x + n, \tag{1}$$

where n represents noise. For this experiment, we use additive Gaussian and magnitude-form Rician noise models during training and evaluation.

D. Restormer Architecture

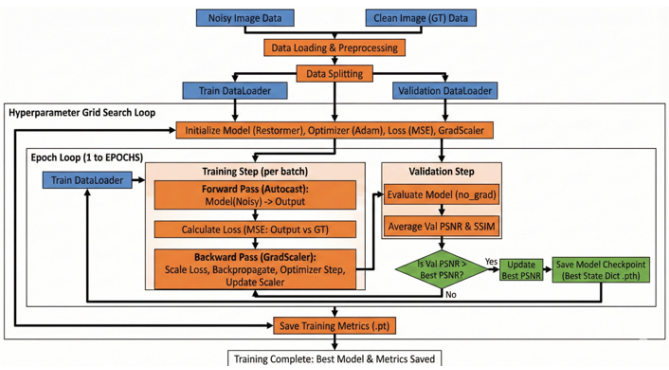

Fig. 1. Data training workflow.

We implemented a Restormer-based U-Net architecture with the following hyperparameters:

- **Channels:** Preprocessing converts images to grayscale. The configuration uses 3 channels to facilitate transfer learning.
- **Base Feature Dimension (`num_feat`):** 128.
- **Levels:** Encoder, bottleneck, decoder.
- **Depth:** $[2, 2, 4, 8]$ blocks across levels.
- **Attention Heads (`num_heads`):** Increases with depth.

E. Training Environment

The model was implemented in PyTorch and trained on an NVIDIA RTX PRO 6000 Blackwell server with PyTorch 2.8-2204 on OpenBayes[1].

F. Data Preprocessing

Source Data
10,000 high-resolution open-source images (Flickr/ImageNet) were used as proxy training data, converted to grayscale, plus 3,000 augmented images. Images were split into training/-validation/test as 80%/10%/10% (Table I). All images were resized to 448×448 and normalized to $[0, 1]$. Quantitative

[1]OpenBayes https://openbayes.com/

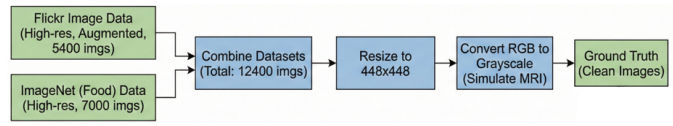

Fig. 2. Data loading pipeline.

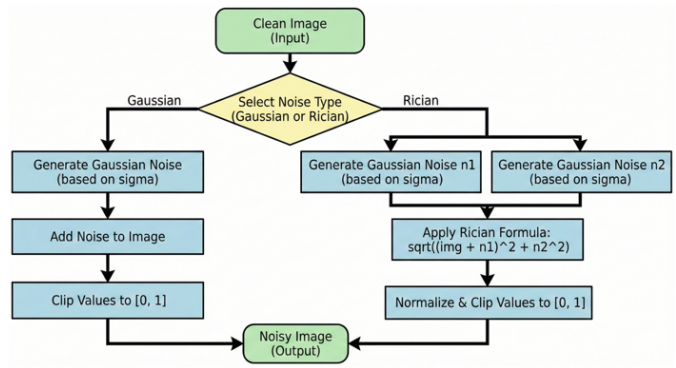

Fig. 3. Data processing and noise injection.

PSNR/SSIM are computed over the validation set ($N_{\text{val}} = 1{,}350$).
Noise Injection
For each image, a target PSNR (dB) is sampled from a clipped normal distribution: $\text{PSNR} \sim \mathcal{N}(26, 2^2)$ clipped to $[22, 30]$. The noise standard deviation is computed (normalized intensity $[0, 1]$):

$$\sigma = 10^{-\text{PSNR}/20}, \tag{2}$$

giving $\sigma \in [0.0316, 0.0794]$. Noise type is sampled with a fixed mixture ratio: 70% Gaussian and 30% Rician. Gaussian noise uses $y = x + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma^2)$. Rician noise is generated by:

$$y = \sqrt{(x + n_1)^2 + n_2^2}, \quad n_1, n_2 \sim \mathcal{N}(0, \sigma^2). \tag{3}$$

Normalization
All images were normalized to $[0, 1]$ (or $[-1, 1]$ depending on implementation) to stabilize training.

TABLE I
DATASET SPLIT AND EVALUATION SIZE.

Subset	Count	Fraction
Training	10,800	80%
Validation	1,350	10%
Test	1,350	10%
Total	13,500	100%

G. Encoder–Decoder Pipeline

- **Downsampling:** Spatial resolution is halved and channels doubled ($112 \rightarrow 224 \rightarrow 448$) using 3×3 stride-2 convolution.
- **Upsampling:** ConvTranspose2d layers upsample features.

- **Skip Connections:** Encoder features are concatenated with decoder features; a 1×1 convolution fuses and reduces channels.
- **Global Residual:** $I_{\text{denoised}} = I_{\text{noisy}} + \mathcal{R}(I_{\text{noisy}})$, where $\mathcal{R}$ is the predicted residual noise map.

H. Training Configuration

Loss Function
Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} \|y_i - \hat{y}_i\|^2. \tag{4}$$

Optimizer
AdamW [17].
Learning Rate
1×10^{-4}.
Weight Decay
Standard regularization.
Batch Size
6 (reduced from 8 due to GPU OOM).
Epochs
100.

I. Evaluation Metrics

PSNR
Peak Signal-to-Noise Ratio (dB); higher is better.
SSIM
Structural Similarity Index Measure; range $[0, 1]$, higher is better [18].

J. Threats to Validity

- **Internal Validity (Synthetic Data Bias):** Real MRI noise can be spatially varying and influenced by k-space sampling and reconstruction.
- **External Validity (Generalization):** Training uses fixed resolution and proxy images; performance may degrade under vendor/protocol shift.

IV. Experimental Results

A. Quantitative Result

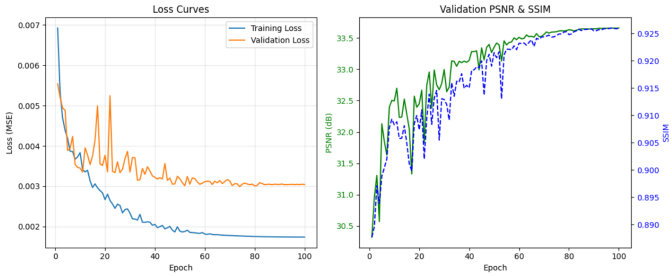

Fig. 4. Learning curves for loss, PSNR, and SSIM over 100 epochs.

Due to budget limits, training was monitored over 100 epochs with checkpoints saved. Fig. 4 shows learning curves for MSE loss and validation PSNR/SSIM.

1) Training Stability and Convergence: Training loss declines and stabilizes around 0.0018, indicating feature learning. Validation loss stabilizes around 0.0030 with no strong "U-shape," suggesting stable generalization.

2) Performance Metrics:

- **Peak Performance:** Epoch 100 achieves PSNR 33.66 dB and SSIM 0.93 on the validation set.
- **Generalization:** Lowest validation loss occurs at Epoch 74 (0.002988); the difference from Epoch 100 is small.

3) Computational Constraints: Training stopped at 100 epochs due to diminishing returns. The Epoch 100 model is selected as the final model based on peak PSNR.

B. Qualitative Result

We compare against BM3D using publicly available Sunnybrook cardiac MRI data[2]. We treat each Sunnybrook slice as a *reference image* and generate a paired noisy input using the same `add_noise` preprocessing pipeline used in our experiments.

1) Dataset-Level Comparison and Runtime: To extend beyond a single example, we evaluated both methods on a Sunnybrook sample set of $N = 148$ images. For each image, we computed PSNR and SSIM with respect to the reference image, and we recorded total execution time for denoising the full set.

As shown in Table II, the proposed Restormer achieves higher average PSNR/SSIM than BM3D on this dataset-level benchmark. In addition, Restormer completes denoising significantly faster. This is expected because BM3D performs per-image patch matching and collaborative filtering operations that are computationally expensive, while Restormer performs a single forward pass per image on the GPU after training.

2) Analysis of Dataset-Level Results: Table II Although the synthetic-noise validation set reports an average PSNR of 33.66 dB, the Sunnybrook benchmark reaches a higher average PSNR (36.98 dB). One possible explanation is that the proxy training/validation images (Flickr/ImageNet) contain more diverse textures, edges, and high-frequency patterns than cardiac MRI, which increases the difficulty of separating "signal" from noise under pixel-wise losses. In contrast, cardiac MRI slices often exhibit smoother intensity regions and more structured boundaries (e.g., organ contours), so the same noise level can be removed with fewer ambiguous high-frequency details.

In addition, the noise added to Sunnybrook images is generated using the same preprocessing pipeline used during training (Gaussian/Rician mixture with PSNR-controlled sampling). This alignment between the training corruption process and the evaluation corruption process can reduce distribution mismatch and produce stronger metrics. Finally, MRI has a more constrained image prior than natural images: repeated structural patterns across slices allow the restoration model to leverage learned priors more effectively, which can improve SSIM by preserving consistent edges while suppressing

[2]Sunnybrook Cardiac Data https://www.cardiacatlas.org/sunnybrook-cardiac-data/

background noise. Overall, these factors help explain why the model achieves stronger average performance on the Sunnybrook set than on the broader proxy-image validation set, while still demonstrating a consistent advantage over BM3D in both fidelity (PSNR/SSIM) and efficiency (runtime).

TABLE II
DATASET-LEVEL DENOISING PERFORMANCE ON SUNNYBROOK SAMPLE SET ($N = 148$) WITH PAIRED SYNTHETIC NOISE. METRICS ARE AVERAGED OVER ALL IMAGES AND COMPUTED RELATIVE TO EACH REFERENCE IMAGE.

Method	PSNR (dB)	SSIM	Time (s)
Noisy Input	26.60	0.42	–
BM3D	32.38	0.69	360
Proposed Model	**36.98**	**0.89**	**35**

3) Single-Image Case Study: To visually validate the structural differences here are some cases for showing graphically.

1) normal sample case. Fig. 5 shows a representative example from the Sunnybrook dataset. BM3D suppresses noise but tends to over-smooth and reduce fine textures. The proposed model better preserves edges and subtle structure, consistent with Restormer's channel-wise attention (MDTA).

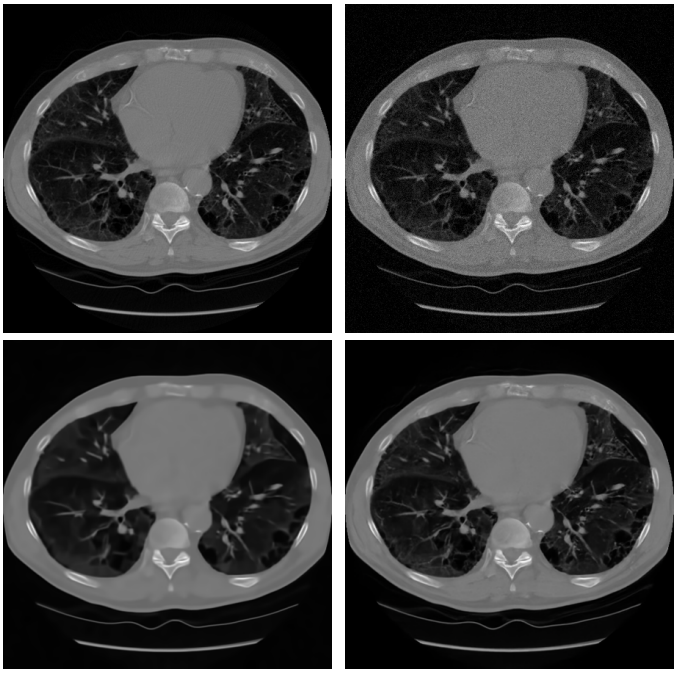

Fig. 5. Single-image example from Sunnybrook. Top: (a) reference image, (b) synthetic noisy input. Bottom: (c) BM3D output, (d) proposed model output. PSNR/SSIM are computed relative to (a).

On this representative slice, the noisy input starts at 28.10 dB. BM3D improves PSNR to 34.58 dB but introduces noticeable over-smoothing, which reduces fine textures and weakens subtle boundary contrast. The proposed model achieves a higher PSNR of 37.81 dB while preserving sharper anatomical edges. This result is consistent with the dataset-level trend in Table II: the learned restoration model can remove noise more effectively than fixed patch-based filtering, and it better maintains structural detail rather than trading it for stronger smoothing.

2) Best-contrast case. Fig. 6 shows an example where BM3D exhibits strong over-smoothing under a difficult noise condition, while the proposed model preserves contrast and fine structure. The noisy input starts at 21.24 dB. BM3D provides only a limited improvement to 23.31 dB, whereas the proposed model reaches 34.27 dB.

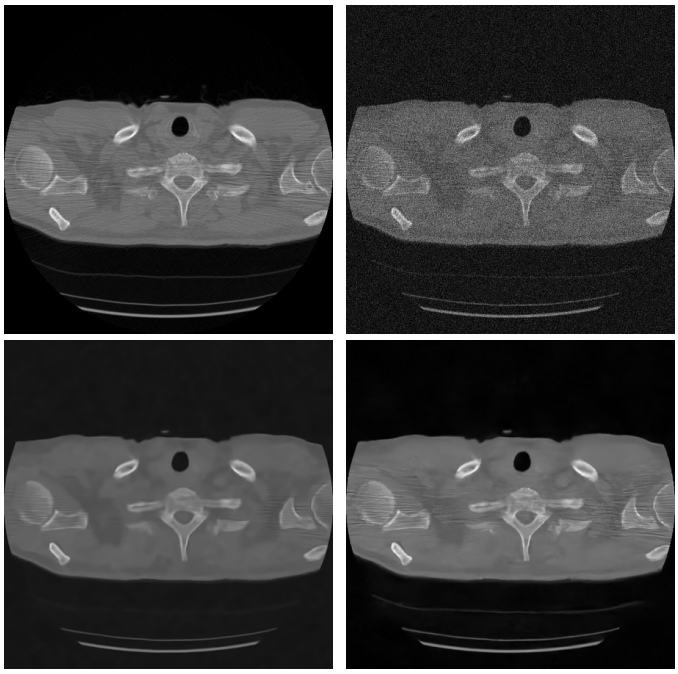

Fig. 6. Best-contrast case. Top: (a) reference image, (b) synthetic noisy input. Bottom: (c) BM3D output and (d) proposed model output. PSNR values are computed relative to (a).

A key reason for this large gap is BM3D's patch-based averaging behavior [4]. When noise is strong, BM3D has difficulty separating subtle tissue texture from noise grain. As a conservative outcome, it removes both, producing a "plastic" appearance with blurred boundaries and reduced internal texture. In contrast, the proposed model is trained to map noisy inputs to clean targets and therefore learns medical-image priors that favor anatomically consistent edges and contrast [12]. In addition, Restormer can use broader contextual cues through attention, which helps reconstruct consistent structures even when local regions are heavily corrupted.

3) Failure case. Fig. 7 shows a case where the proposed model provides only a small improvement over BM3D. The noisy input starts at 29.28 dB. BM3D reaches 31.86 dB, while the proposed model reaches 32.77 dB.

Potential reason is that this MRI image is dominated by large smooth regions (e.g., relatively uniform organ tissue). BM3D performs patch matching and averaging, which is well-suited to denoising piecewise-smooth content. In this scenario, BM3D's tendency to smooth aggressively can reduce pixel-wise error in broad homogeneous areas, producing a strong

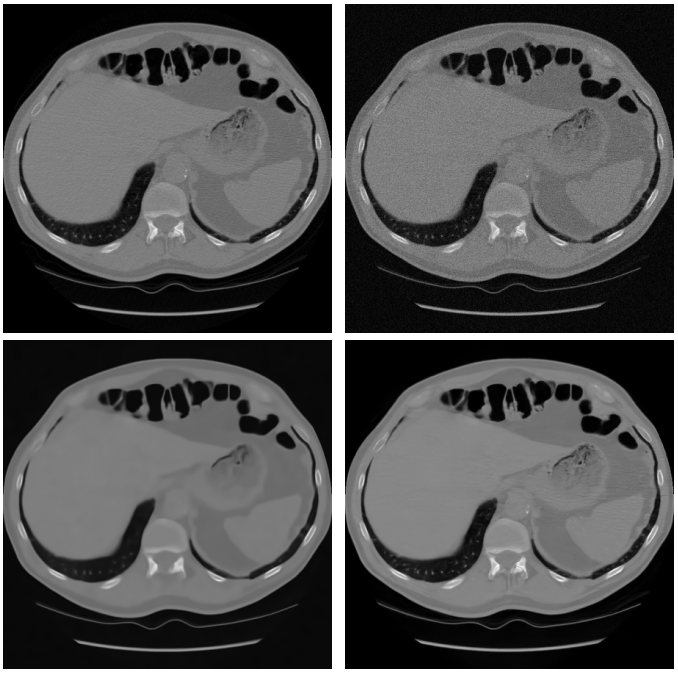

Fig. 7. Best-contrast case. Top: (a) reference image, (b) synthetic noisy input. Bottom: (c) BM3D output and (d) proposed model output. PSNR values are computed relative to (a).

PSNR baseline. In contrast, the proposed model aims to preserve sharper boundaries and subtle structures. While this is often visually preferable, PSNR is based on pixel-wise mean squared error, so sharp but slightly misaligned edges can be penalized [19]. Finally, because the BM3D baseline is already high here, the remaining headroom for PSNR improvement is limited; improvements from 30–32 dB typically require much finer pixel-level precision than improvements at very low PSNR.

Therefore, this case should be interpreted as a *limited-gain* example rather than a denoising failure. The proposed model still preserves structural boundaries more clearly, whereas BM3D may produce a waxy appearance that can suppress subtle details that may be clinically relevant.

Summary of the case study: Across the three single-image examples, the proposed model consistently preserves sharper anatomical boundaries and more realistic texture than BM3D. In the normal case (Fig. 5), Restormer achieves higher PSNR while avoiding BM3D over-smoothing. In the best-contrast case (Fig. 6), BM3D collapses fine structure under heavy noise, the proposed model restores contrast and detail with a large PSNR gain. In the limited-gain case (Fig. 7), BM3D performs relatively well on smooth regions, so PSNR improvement is smaller; however, the proposed model still yields clearer edges. Overall, these cases support that the learned model trades less structural detail for smoothing compared with BM3D, especially when noise is strong or textures are important.

V. Conclusions

This project aimed to improve cardiac MRI denoising while preserving anatomical structure. With reference to the research objectives, the following conclusions are drawn:

- **RO1:** Mixed Gaussian/Rician noise injection provides a practical approximation of MRI-like noise behavior, and the Sunnybrook case study suggests structure can be preserved on real anatomy.
- **RO2:** A Restormer-based restoration network was implemented and adapted for cardiac MRI denoising, preserving structure better than BM3D in the case study.
- **RO3:** The model achieves validation PSNR 33.66 dB and SSIM 0.93 at epoch 100 under the synthetic-noise protocol.
- **RO4:** Training converges stably under GPU constraints (feature width 128, batch size 6), supporting feasibility on affordable settings [20].

A. Answers to Research Questions and Hypotheses

- **RQ1/H1:** The Sunnybrook single-image case study shows higher PSNR/SSIM than BM3D on that example. BM3D was not evaluated as a dataset-level average on the full split, so H1 is supported only at the case-study level.
- **RQ2/H2:** Mixed-noise training improves qualitative structure preservation in the Sunnybrook case study, consistent with H2; broader validation is needed.

B. Lessons Learnt

1) **Memory limits are a primary constraint:** Training memory is dominated by feature maps and activations; channel width increases VRAM usage substantially.
2) **Iteration is required for feasibility:** OOM mitigation required repeated batch size and pipeline adjustments.
3) **Evaluation must match the target:** Metrics must be reported with a clearly stated evaluation protocol; qualitative inspection remains necessary for anatomical fidelity.

C. Future Work

1) **Real-world validation:** Evaluate on raw k-space and real clinical scanners to capture motion/coil effects.
2) **3D processing:** Extend from 2D slices to pseudo-3D or 3D architectures.
3) **Cross-vendor generalization:** Test robustness across vendors/protocols and apply domain adaptation if needed.
4) **Training with real MRI datasets:** The current model was trained using simulated noise on proxy images, yet it already shows strong potential for improving MRI denoising in resource-limited clinics. Future work should train and fine-tune Restormer on real paired or self-supervised MRI datasets to better capture scanner-specific noise characteristics and further improve clinical performance.

D. Recall

Overall, the results in this report suggest that a Restormer-based denoising pipeline is a credible and practical direction for improving MRI image quality under limited data and hardware constraints. While broader clinical validation is still required, the observed gains over BM3D and the strong runtime advantage indicate that this approach is worth further development toward real-world cardiac MRI denoising in clinics usage.

VI. Reproducibility

For reproducibility, ensure paths and folder names are set correctly.

A. Data Processing

Create the following folders:

- `GT`
- `GT_noise`
- `denoise`

The `GT` folder saves reference images, `GT_noise` saves noisy images generated by `add_noise.py`, and `denoise` saves denoised outputs. Execute scripts in this order:

1) `img_to_grey.py`
2) `resize.py`
3) `add_noise.py`

B. Training

Use:

- `Restormer.ipynb` (training / resume from checkpoint)
- `Restormer_analysis.ipynb` (analysis of saved `.pt` files)

C. Denoising

Use:

- `denoise_restormer.py`

D. Benchmarking

Use:

- `benchmarking.py`

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